

Food Pantry - Database Schema

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Description:	This document provides a structured overview of the Food Pantry application's database schema, including entity definitions, relationships, attributes, and cardinalities to support application functionality and data integrity.

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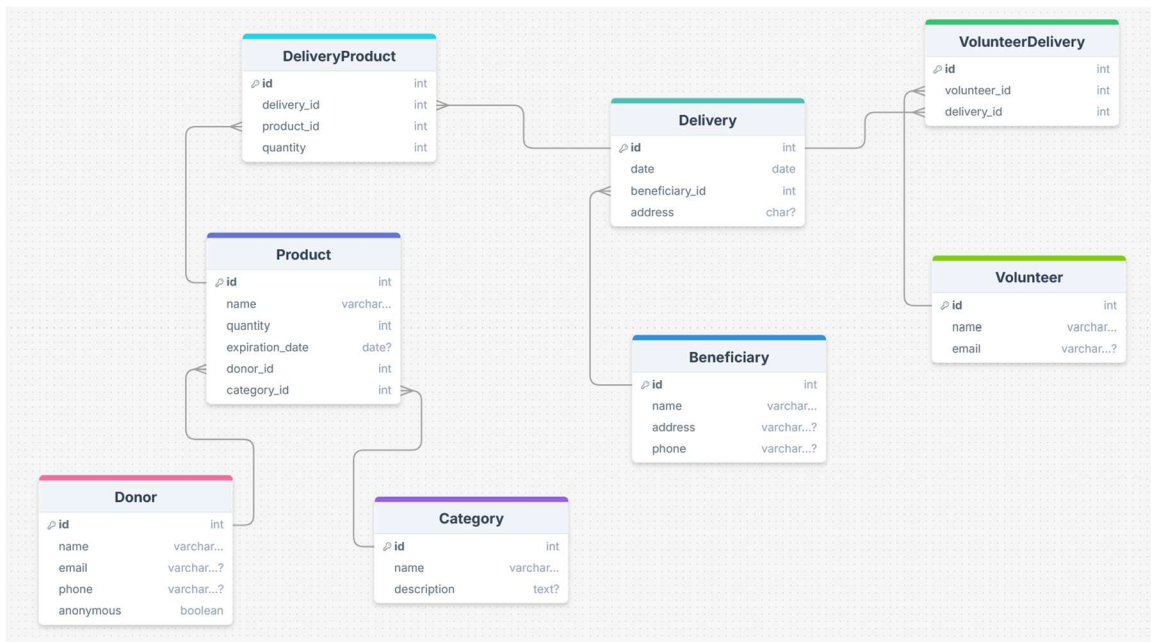
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1. Introduction

This document describes the relational database schema of the **Food Pantry** system, which supports operations such as product distribution, volunteer coordination, donor registration, and beneficiary management.

2. ER Diagram

The following image represents the Entity-Relationship (ER) model for the Food Pantry system.



3. Entities Overview

Donor

Stores information about the people or organizations that donate food.

Product

Represents food items available for delivery.

Category

Classifies products into categories like fruits, vegetables, etc.

Beneficiary

Stores people who receive food deliveries.

Delivery

Links beneficiaries to deliveries with date and address.

DeliveryProduct

Intermediate table linking deliveries and products (many-to-many).

Volunteer

Stores volunteers who assist with deliveries.

VolunteerDelivery

Links volunteers to deliveries (many-to-many).

4. Relationships and Cardinality

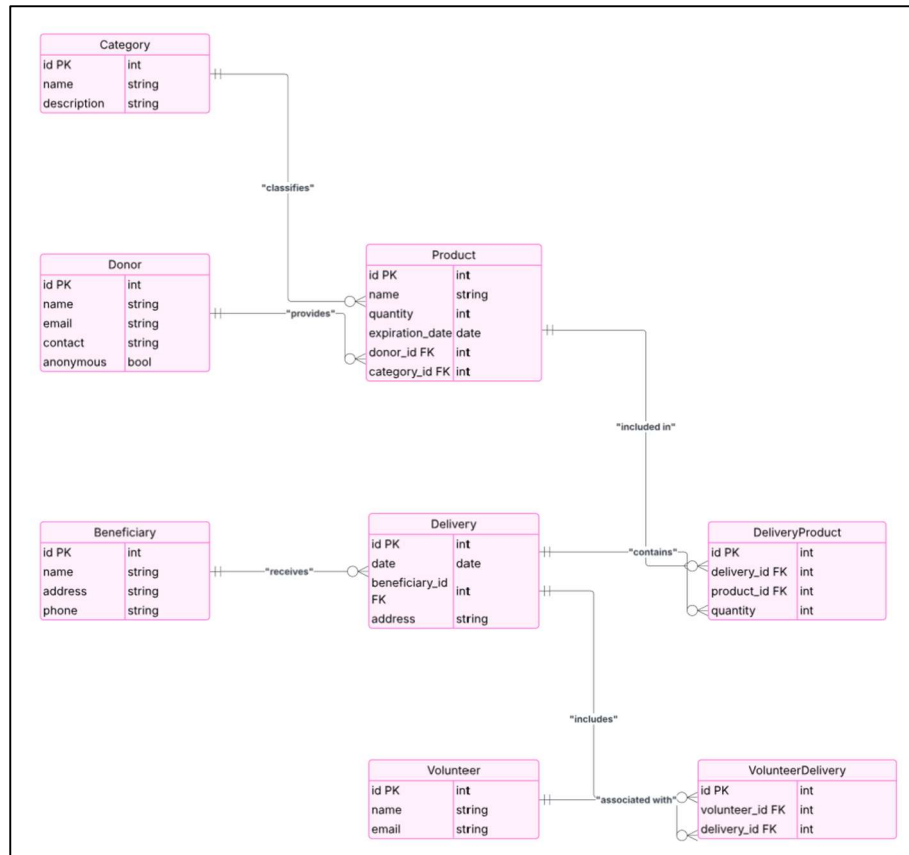
- A Donor can donate multiple Products (1:N).
- A Category can group multiple Products (1:N).
- A Product can belong to one Donor and one Category.
- A Beneficiary can receive multiple Deliveries (1:N).
- A Delivery can contain multiple Products through DeliveryProduct (N:M).
- A Volunteer can participate in multiple Deliveries through VolunteerDelivery (N:M).

Relationship	Cardinality	Description
Donor → Product	1:N	A Donor can donate multiple Products (1:N).
Category → Product	1:N	A Category can group multiple Products (1:N).
Beneficiary → Delivery	1:N	A Beneficiary can receive multiple Deliveries (1:N).
Delivery ↔ Product	N:M (via DeliveryProduct)	A Delivery can contain multiple Products through DeliveryProduct (N:M).
Delivery ↔ Volunteer	N:M (via VolunteerDelivery)	A Volunteer can participate in multiple Deliveries through VolunteerDelivery (N:M).

Extra relationships and restrictions implemented in the code logic:

Entity 1	Relationship	Entity 2	Cardinality	Description
Donor	donates	Product	1:N (one to many)	A donor can donate multiple products, but each product is donated by one donor.
Category	classifies	Product	1:N (one to many)	A category can group multiple products, but each product belongs to one category.
Product	included in	DeliveryProduct	1:N (one to many)	A product can be included in multiple deliveries via the intermediate table.
Delivery	contains	DeliveryProduct	1:N (one to many)	A delivery can contain many products.
Beneficiary	receives	Delivery	1:N (one to many)	A beneficiary can receive many deliveries, each delivery is assigned to one beneficiary.
Delivery	performed by	VolunteerDelivery	1:N (one to many)	A delivery can be performed by multiple

				volunteers (through intermediate table).
Volunteer	participates in	VolunteerDelivery	1:N (one to many)	A volunteer can participate in multiple deliveries.



Schema with cardinality

5. Attribute Tables

Donor

-- Table storing donor information (anonymous or identified)

Attribute	Type / Constraint
id	int (PK)
name	Varchar(180)
email	Varchar(180)
contact	Varchar(180)
anonymous	Boolean (default=false)

Product

-- Table storing donated products

Attribute	Type / Constraint
id	int (PK)
name	Varchar(180)
quantity	int
expiration_date	date
donor_id	int (FK)
category_id	int (FK)

Category

-- Table classifying product categories

-- This table is used to categorize products for better organization and searchability

-- It allows products to be grouped by type, such as "Canned Goods", "Fresh"

Attribute	Type / Constraint
-----------	-------------------

id	int (PK)
name	Varchar(180)
description	Text(blank=True, Null=True)

Beneficiary

-- Table storing information about product beneficiaries

Attribute	Type / Constraint
id	int (PK)
name	varchar
address	varchar?
phone	varchar?

Delivery

-- Table recording each delivery to a beneficiary

Attribute	Type / Constraint
id	int (PK)
date	date
beneficiary_id	int (FK)
address	char?

DeliveryProduct

-- Table recording each delivery to a beneficiary

Attribute	Type / Constraint
id	int (PK)
delivery_id	int (FK)
product_id	int (FK)

quantity	int
----------	-----

Volunteer

-- Table for volunteers involved in logistics

Attribute	Type / Constraint
id	int (PK)
name	Varchar(100)
email	Varchar(200)

VolunteerDelivery

-- Table for volunteers involved in logistics

Attribute	Type / Constraint
id	int (PK)
volunteer_id	int (FK)
delivery_id	int (FK)

6. SQL code

Here, it provides the SQL code, but the code is implemented in Django models.py of every api (donors_api, products_api, volunteers_api, deliveries_api,...).

```
-- Table storing donor information (anonymous or identified)
```

```
CREATE TABLE "Donor"(
    "id" INTEGER NOT NULL,
    "name" VARCHAR(100) NOT NULL,
    "email" VARCHAR(100) NULL,
    "phone" VARCHAR(15) NULL,
    "anonymous" BOOLEAN NOT NULL
```

```
);  
  
ALTER TABLE  
  
    "Donor" ADD PRIMARY KEY("id");  
  
-- Table classifying product categories  
-- This table is used to categorize products for better organization and  
searchability  
-- It allows products to be grouped by type, such as "Canned Goods", "Fresh  
CREATE TABLE "Category"(  
    "id" INTEGER NOT NULL,  
    "name" VARCHAR(100) NOT NULL,  
    "description" TEXT NULL  
);  
  
ALTER TABLE  
  
    "Category" ADD PRIMARY KEY("id");  
  
-- Table storing donated products  
CREATE TABLE "Product"(  
    "id" INTEGER NOT NULL,  
    "name" VARCHAR(100) NOT NULL,  
    "quantity" INTEGER NOT NULL,  
    "expiration_date" DATE NULL,  
    "donor_id" INTEGER NOT NULL,  
    "category_id" INTEGER NOT NULL  
);  
  
ALTER TABLE  
  
    "Product" ADD PRIMARY KEY("id");
```

-- Table storing information about product beneficiaries

```
CREATE TABLE "Beneficiary"(  
    "id" INTEGER NOT NULL,  
    "name" VARCHAR(100) NOT NULL,  
    "address" VARCHAR(255) NULL,  
    "phone" VARCHAR(15) NULL  
);  
  
ALTER TABLE  
    "Beneficiary" ADD PRIMARY KEY("id");
```

-- Table recording each delivery to a beneficiary

```
CREATE TABLE "Delivery"(  
    "id" INTEGER NOT NULL,  
    "date" DATE NOT NULL,  
    "beneficiary_id" INTEGER NOT NULL,  
    "address" CHAR(255) NULL  
);  
  
ALTER TABLE  
    "Delivery" ADD PRIMARY KEY("id");
```

-- Table recording each delivery to a beneficiary

```
CREATE TABLE "DeliveryProduct"(  
    "id" INTEGER NOT NULL,  
    "delivery_id" INTEGER NOT NULL,  
    "product_id" INTEGER NOT NULL,
```

```
        "quantity" INTEGER NOT NULL
    );

ALTER TABLE

    "DeliveryProduct" ADD PRIMARY KEY("id");

-- Table for volunteers involved in logistics
CREATE TABLE "Volunteer"(
    "id" INTEGER NOT NULL,
    "name" VARCHAR(100) NOT NULL,
    "email" VARCHAR(100) NULL
);

ALTER TABLE

    "Volunteer" ADD PRIMARY KEY("id");

-- Table for volunteers involved in logistics
CREATE TABLE "VolunteerDelivery"(
    "id" INTEGER NOT NULL,
    "volunteer_id" INTEGER NOT NULL,
    "delivery_id" INTEGER NOT NULL
);

ALTER TABLE

    "VolunteerDelivery" ADD PRIMARY KEY("id");

ALTER TABLE

    "DeliveryProduct" ADD CONSTRAINT "deliveryproduct_product_id_foreign"
FOREIGN KEY("product_id") REFERENCES "Product"("id");

ALTER TABLE
```

```
"VolunteerDelivery" ADD CONSTRAINT
"volunteerdelivery_volunteer_id_foreign" FOREIGN KEY("volunteer_id")
REFERENCES "Volunteer"("id");

ALTER TABLE

"Delivery" ADD CONSTRAINT "delivery_beneficiary_id_foreign" FOREIGN
KEY("beneficiary_id") REFERENCES "Beneficiary"("id");

ALTER TABLE

"VolunteerDelivery" ADD CONSTRAINT "volunteerdelivery_delivery_id_foreign"
FOREIGN KEY("delivery_id") REFERENCES "Delivery"("id");

ALTER TABLE

"DeliveryProduct" ADD CONSTRAINT "deliveryproduct_delivery_id_foreign"
FOREIGN KEY("delivery_id") REFERENCES "Delivery"("id");

ALTER TABLE

"Product" ADD CONSTRAINT "product_donor_id_foreign" FOREIGN
KEY("donor_id") REFERENCES "Donor"("id");

ALTER TABLE

"Product" ADD CONSTRAINT "product_category_id_foreign" FOREIGN
KEY("category_id") REFERENCES "Category"("id");
```